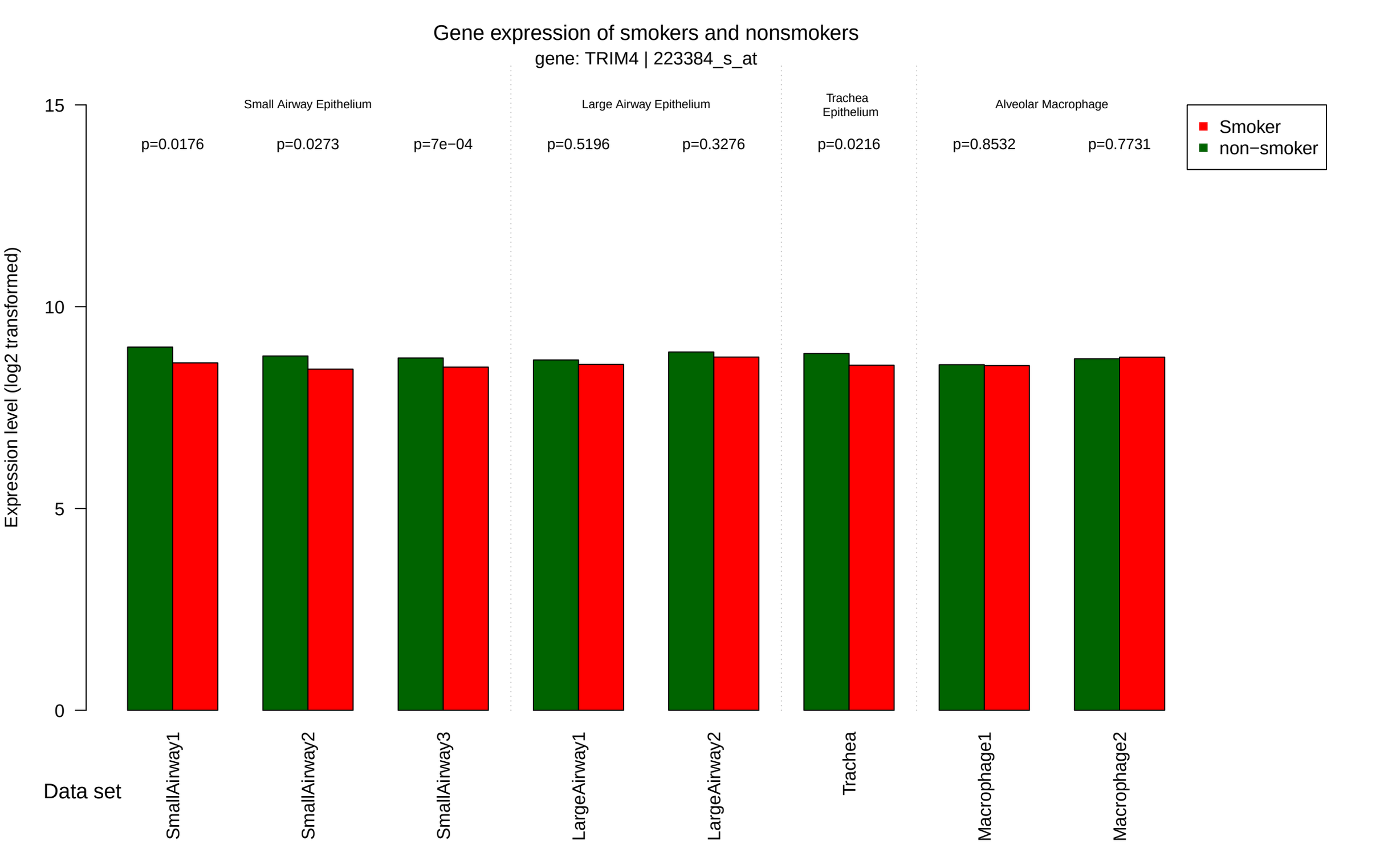
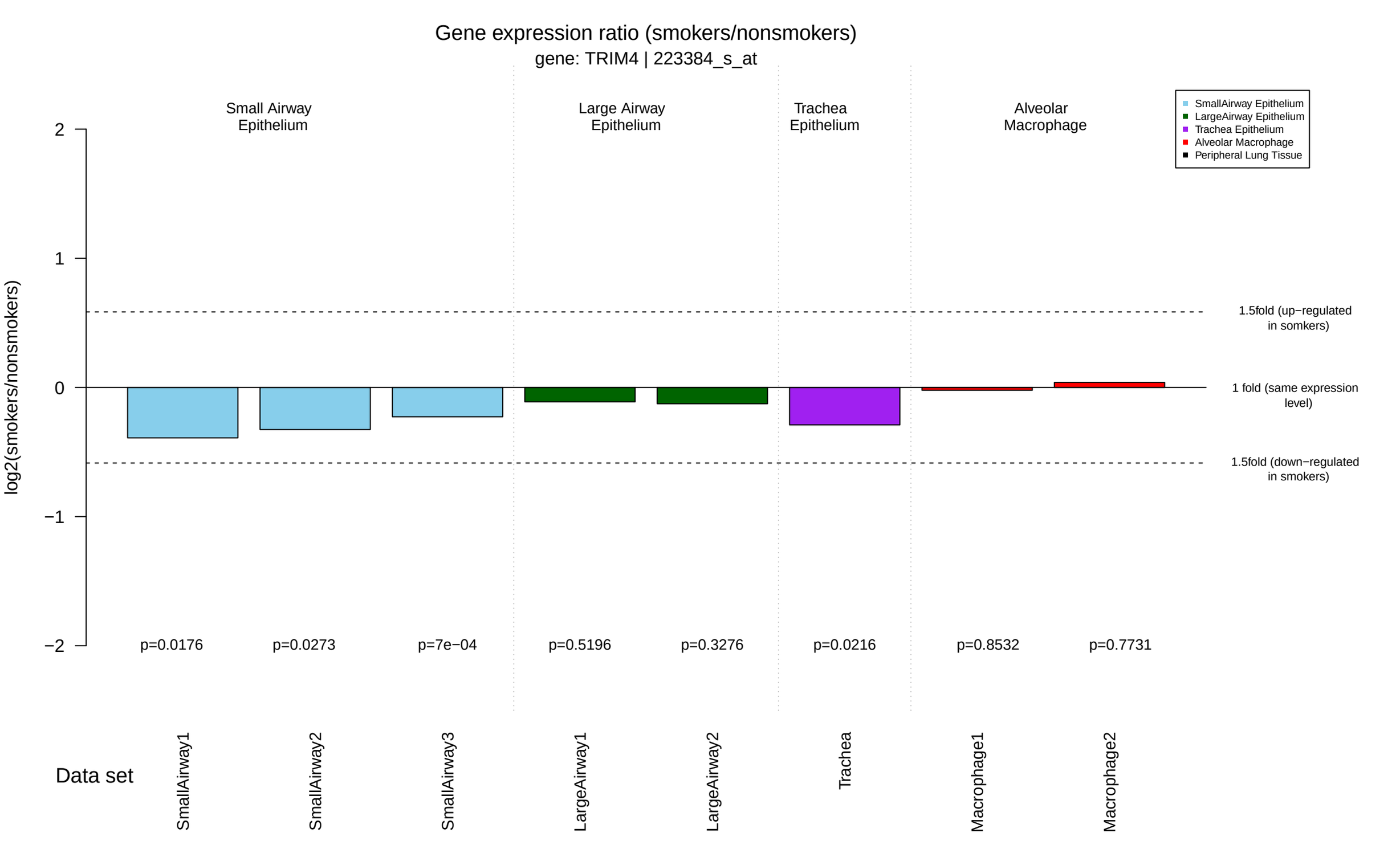
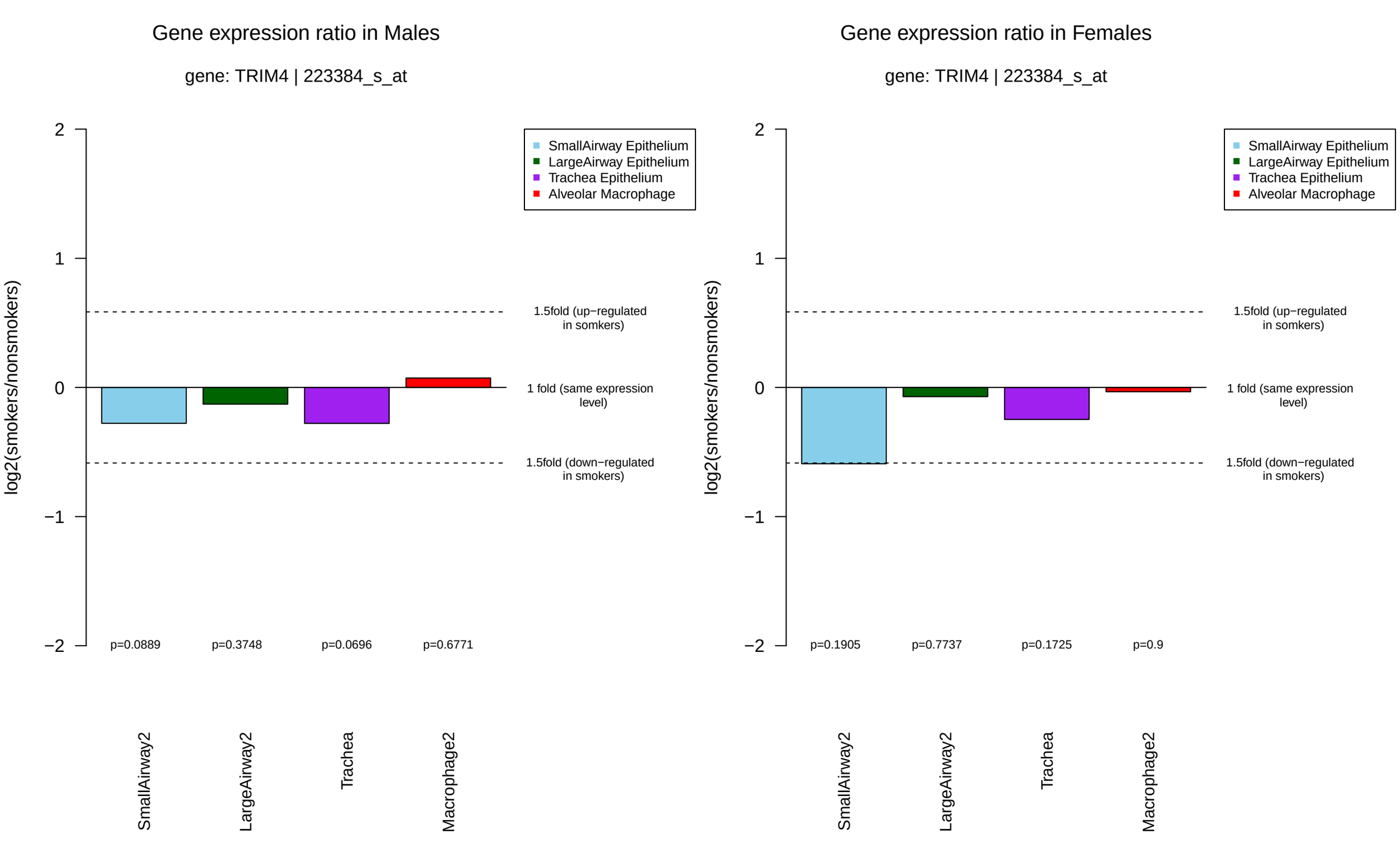
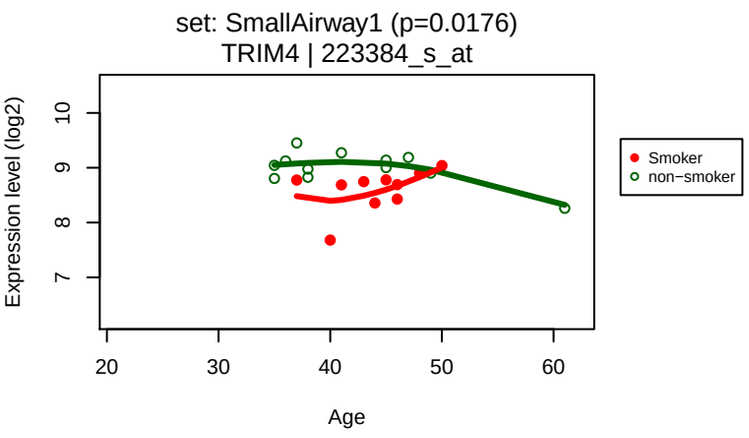
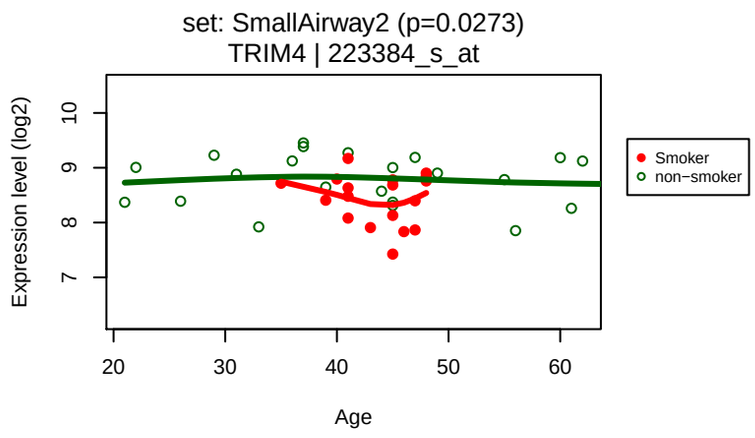
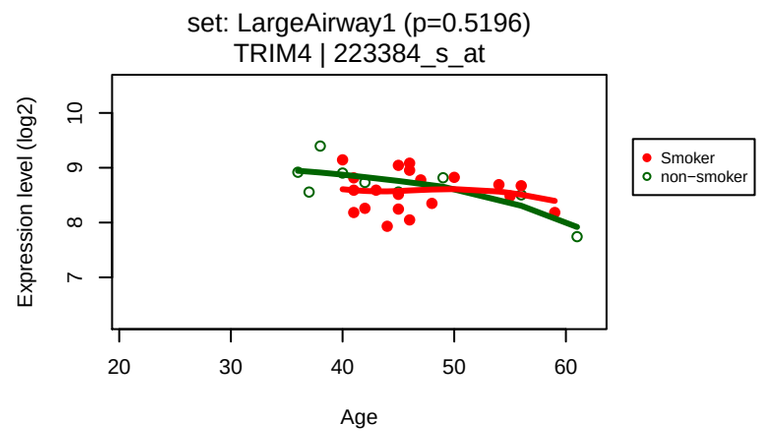

Gene expression of smokers and nonsmokers
gene: TRIM4 | 223384_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0176
p=0.0273
p=7e−04
p=0.5196
p=0.3276
p=0.0216
p=0.8532
p=0.7731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

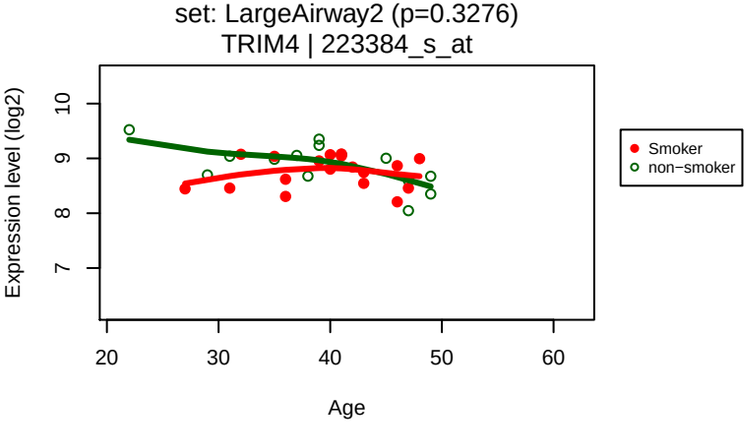

Gene expression ratio (smokers/nonsmokers)
gene: TRIM4 | 223384_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0176
p=0.0273
p=7e−04
p=0.5196
p=0.3276
p=0.0216
p=0.8532
p=0.7731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

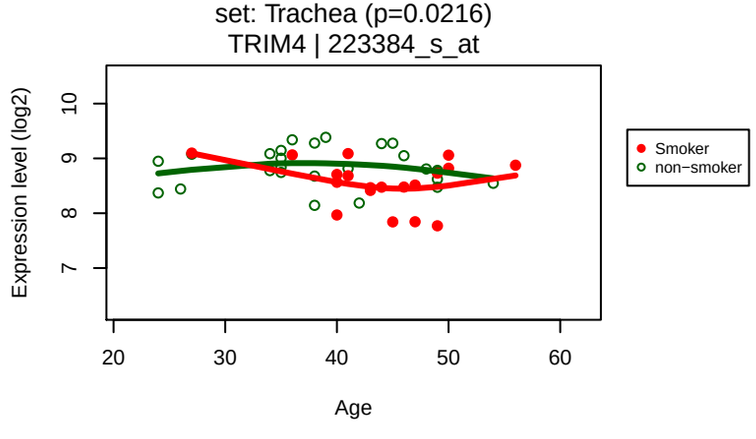

Gene expression ratio in Males
gene: TRIM4 | 223384_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0889
p=0.3748
p=0.0696
p=0.6771
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRIM4 | 223384_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1905
p=0.7737
p=0.1725
p=0.9
SmallAirway2
LargeAirway2
Trachea
Macrophage2

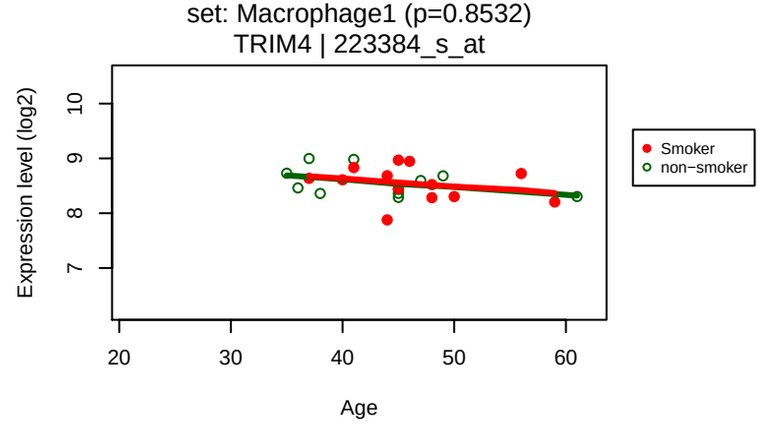

set: SmallAirway1 (p=0.0176)
TRIM4 | 223384_s_at
Expression level (log2)
Age
Smoker
non−smoker

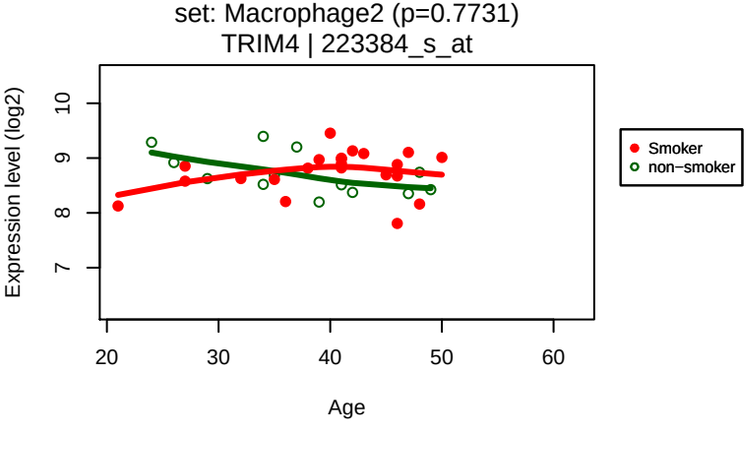

set: SmallAirway2 (p=0.0273)
TRIM4 | 223384_s_at
Expression level (log2)
Age
Smoker
non−smoker

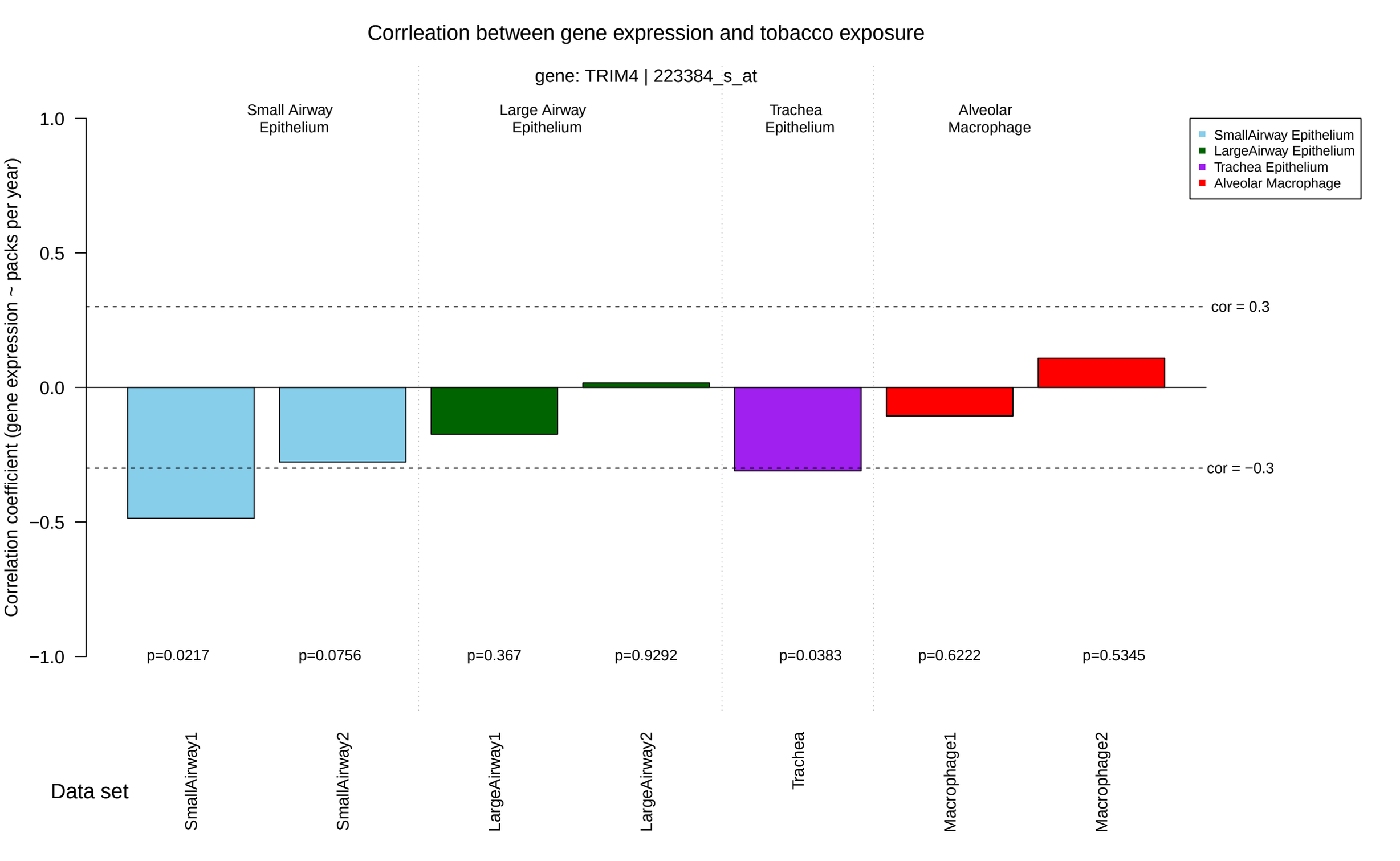

set: LargeAirway1 (p=0.5196)
TRIM4 | 223384_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3276)
TRIM4 | 223384_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0216)
TRIM4 | 223384_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8532)
TRIM4 | 223384_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7731)
TRIM4 | 223384_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRIM4 | 223384_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0217
p=0.0756
p=0.367
p=0.9292
p=0.0383
p=0.6222
p=0.5345
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2